Safety and short-term outcomes of lecanemab for Alzheimer's disease in China: a multicentre study

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Abstract

Lecanemab is a newly approved monoclonal antibody targeting amyloid plaques for the treatment of early Alzheimer's disease. This study aimed to evaluate the safety and short-term biomarkers and cognition changes of lecanemab in Chinese clinical practice.

This multicenter real-world study involved patients receiving lecanemab treatment across seven hospitals in China. Patients underwent comprehensive assessments before treatment. Lecanemab was administered via intravenous infusion every 2 weeks. Treatment-related symptoms were monitored through self-report, and amyloid-related imaging abnormalities were assessed via magnetic resonance imaging. Amyloid and tau biomarker changes were measured using positron emission tomography imaging or plasma testing. Follow-up cognitive assessments were evaluated after 6 months of treatment. Short-term outcomes were analyzed using linear mixed-effect models, without an untreated control group.

A total of 407 patients who received at least one lecanemab infusion were involved in this study, with a mean follow-up time of 5.6±3.39 months. The mean age was 68.08 years, with 67.57% of patients being female. Of the participants, 56.51% were APOE ε4 carriers, and 83.19% were at biological stage C. During the study period, 22.22% of the patients experienced treatment-

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related symptoms, and 12.15% developed at least one amyloid-related imaging abnormality. Only four symptomatic and seven severe amyloid-related imaging abnormality cases were reported. APOE ε4 status was not related to adverse events in the Chinese population. Patients with a higher number of microhemorrhages at baseline were more likely to develop adverse events. No significant differences in adverse events were observed between the moderate Alzheimer's disease dementia group and the mild cognitive impairment group. By the end of the research period, 9.38% of the patients withdrew from lecanemab. After 6 months of treatment, favourable short-term outcomes in biomarkers and stable cognitive function were observed.

This study demonstrates that lecanemab treatment is feasible and well-tolerated among the Chinese population, with lower rates of adverse events and favourable short-term outcomes observed. Administration of lecanemab in moderate AD dementia population was relatively safe and further studies are warranted.

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Introduction

Alzheimer's disease (AD) is the predominant etiology of dementia, currently affects over 55 million individuals worldwide, with prevalence projected to approach 152 million cases by 2050[1]. Its defining neuropathological features are extracellular β-amyloid (Aβ) plaque deposition and intraneuronal hyperphosphorylated tau aggregation[2]. Lecanemab, a humanized monoclonal antibody targeting soluble Aβ protofibrils, has emerged as a significant therapeutic breakthrough[3]. In the phase III Clarity AD trial, lecanemab has slowed cognitive decline by 27% in patients with early AD [4], leading to regulatory approval for use in mild cognitive impairment (MCI) and mild AD dementia in multiple countries.

Although lecanemab, as the first disease-modifying therapy, has demonstrated clinical benefits, its translation into real-world practice requires further consideration. In East Asian populations, where APOE ε4 confers a greater risk for AD and cerebral microbleeds are more prevalent[5,6], there is concern that this group may be more prone to amyloid-related imaging abnormalities (ARIA) risk[7,8]. While the Clarity AD trial in Asian subjects[9] reported a lower incidence of

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adverse events (AEs), including infusion-related reactions and ARIA, than the whole group, these findings are derived from rigorously screened cohorts under controlled conditions, limiting their generalizability to routine clinical practice. The safety of lecanemab treatment in clinical practice should be further examined.

The differences in cognitive function cut-off criteria and education levels between Asian and Western populations[10] also pose challenges for accurately determining the clinical stage. Patient eligibility may be affected by simply evaluating the patients' cognitive function to determine whether the medication can be administered. Furthermore, as the understanding and diagnosis of AD is shifting to a biologically based framework[11], optimal biomarker staging in real-world settings may complicate the management strategies and efficacy prediction of lecanemab.

To address these evidence gaps, we established China's first multicenter study of patients treated with lecanemab, with the following primary objectives: (1) to examine the safety and short-term changes of biomarkers and cognition with lecanemab in the Chinese population in a real-world clinical setting; (2) to identify potential risk factors associated with the incidence of AEs; and (3) to explore the use of lecanemab across a broader patient population. This investigation aims to provide an empirical foundation for the precision implementation of lecanemab across diverse healthcare systems.

Materials and methods

Study Population

This multi-center study involved patients from seven hospitals in China, including Huashan Hospital Affiliated to Fudan University, First Affiliated Hospital Zhejiang University School of Medicine, Fujian Medical University Union Hospital, The First Affiliated Hospital of Wenzhou Medical University, The Second Affiliated Hospital of Soochow University, Nanjing First Hospital, and Qingdao Municipal Hospital. These centers provide a nationally representative sample spanning Shanghai, Zhejiang, Jiangsu (East China), Fujian (South China), and Shandong (North China). Consecutive patients who received at least one lecanemab infusion ordered by these centers from June 26, 2024, to May 26, 2025, were included in this study. All participants

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underwent a comprehensive diagnostic evaluation, including medical history, clinical examination, neuropsychological assessments, routine laboratory analyses, brain MRI, and biomarker testing. The diagnosis and staging of AD were based on the latest criteria[11]. After a thorough examination and diagnosis, patients identified as potential candidates for lecanemab referring to the appropriate use recommendations[12] and interested in the treatment were included in the unified management. The study protocol received approval from the Ethics Committee of each center. Informed consent was obtained from all participants or their legally authorized representatives.

Treatment Workflow

This is a prospectively designed cohort study with consecutive patients initiating lecanemab from each center, the inclusion workflow is shown in Fig. 1A. Candidates for lecanemab treatment were further classified into stages of AD based on their severity of cognitive impairment as rated by the Clinical Dementia Rating (CDR). All candidates had a diagnosis of symptomatic AD with a CDR of 0.5 (MCI), 1 (mild dementia), or 2 (moderate dementia). Most candidates had a Mini-Mental State Examination (MMSE) score of 20 or higher. Activities of Daily Living (ADL) scales was additionally assessed. The number of APOE ε4 alleles was tested in each center using standard laboratory techniques. For patients with conditions that would have excluded them from treatment, including moderate dementia and those taking anticoagulant medications, they were informed by the doctor of the reasons why they were not suitable for lecanemab and of the serious side effects. For patients who insisted on lecanemab treatment, the neurology specialist conducted a more comprehensive evaluation and engaged in in-depth discussions with patients and their families.

Lecanemab was administered via intravenous infusion every 2 weeks at infusion centers. Patients from each center were suggested to report all symptoms that occurred during their treatment period and complete monitoring MRI, biomarker testing, and cognitive assessments. Structured checklists for symptom capture were distributed to each participating center prior to the initiation of the study. All AEs considered to be related to lecanemab treatment were recorded by self-reported symptoms or by monitoring MRI. Treatment-related symptom was defined as all

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symptoms related to lecanemab, and were categorized by System Organ Class (SOCs) from the Medical Dictionary of Adverse Drug Reactions and graded according to the Common Terminology Criteria for Adverse Events[13]. Infusion-related reaction (IRR) was defined as symptoms during or after lecanemab infusion (within 24 hours) due to the drugs, procedures, or individual differences. The severity of symptoms was categorized as mild (does not affect daily life), moderate (affecting daily life and requiring intervention treatment), and severe (requiring strict monitoring and treatment). The patients were followed till the latest infusion or withdrawal from the treatment. For the handling of AEs, such as prophylactic treatment, suspension and resumption of lecanemab, each center independently made the determination based on clinical practice under the guidance of the drug instructions[14]. Patients underwent monitoring MRIs before the fifth, seventh, and fourteenth infusions. Urgent monitoring MRI was performed on patients with sudden symptoms. For patients who agreed to regularly monitor biomarker changes, plasma was collected at baseline, 3 months, and 6 months after treatment initiation, and amyloid and tau PET was conducted at baseline and 6 months after treatment initiation. Follow-up cognitive assessments were conducted 6 months after treatment initiation by the same rater. All collected follow-up biomarkers and cognitive assessment results were included in the analysis of changes after short-term treatment.

MRI

All patients underwent a pretreatment brain MRI that consisted of T1, T2, fluid-attenuated inversion recovery (FLAIR), diffusion weighted imaging (DWI), and susceptibility weighted imaging (SWI). Radiological findings including the severity of white matter disease assessed by the Fazekas score, the presence and size of infarcts, the number of microhemorrhages, and superficial siderosis were evaluated. Follow-up MRI scans were performed to assess the presence of amyloid-related imaging abnormalities with edema/effusion (ARIA-E) and hemorrhage/hemosiderin deposition (ARIA-H) including a sagittal T1, axial DWI and T2, FLAIR, and SWI. MRI sequences conducted outside centers with different scanner strengths were reviewed by a neurology specialist to determine the presence of ARIA-E and ARIA-H. Monitoring MRI scans were scheduled before the 5th, 7th, and 14th infusion.

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For patients who developed ARIA, the severity of ARIA was categorized consistent with FDA guidelines[14]. Concurrent ARIA-E and ARIA-H were defined as presence of ARIA-E and ARIA-H in the same monitoring period, regardless of whether they are located in the same brain region. Most people underwent a reviewing MRI one month after the occurrence of ARIA. Symptomatic ARIA was defined as the presence of an ARIA finding on regular or urgent MRI in patients presenting with clinical symptoms attributable to ARIA.

PET Imaging

PET scans were performed with the Biograph mCT Flow PET/CT system (Siemens, Erlangen, Germany), uMI 780 PET/CT system, uMI Panorama 35S PET/CT system, uMI 790 PET/MR system (United Imaging Intelligence Co., Ltd., China), and Discovery MI PET/CT system (GE HealthCare, United States) in each center. Either ^{18}F-Flobetapir or ^{18}F-Forbetaben positron emission tomography (PET) scans were acquired and subjected to visual assessment for initial confirmation of Aβ deposition. Tau pathology was performed by ^{18}F-Florzolotau, ^{18}F-Flortaucipir, or ^{18}F-MK6240 PET images. Following the acquisition, the data underwent reconstruction using a three-dimensional ordered-subset expectation maximization algorithm. The images were visually evaluated by two experienced nuclear medicine physicians. The staging of tau PET was based on previous literature[11] and was determined through visual read. Specifically, abnormal amyloid PET patients were categorized into Stage A (initial-no uptake on tau PET), B (early-tau PET uptake restricted to medial-temporal areas), C (intermediate-tau PET uptake in the moderate standardized uptake value ratio [SUVR] range on a neocortical region of interest [ROI]), and D (advanced-tau PET uptake in the high SUVR range in the same neocortical ROI) by tau PET results. Any disagreement between the two readers was resolved by consulting a third senior assessor. The amyloid centiloid was calculated based on the previously published method[15,16]. For tau PET SUVR in the ROIs, the cerebellar gray matter was selected as the reference region. Anatomical ROIs encompassing both cortical and subcortical regions were defined based on the Desikan-Killiany atlas[17].

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Biomarker Testing

The CSF AD biomarker testing was conducted with standard ELISA methods (EUROIMMUN AG, Lübeck, Germany, CE-registered assay). Samples were collected and transported according to the requirements of the kit manufacturers and the Alzheimer's Association's latest guideline[18]. CSF biomarkers including Aβ1-42, Aβ1-40, total tau, and phosphorylated tau 181 were tested. All handling steps were performed according to the manufacturer's instructions.

Blood biomarker testing was performed by single-molecule array technology. Biomarkers including Aβ42, Aβ40, glial fibrillary acidic protein (GFAP), neurofilament light chain (NfL), and P-Tau217 were tested. Single-molecule array technology was performed by the ALZpath P-Tau217 Advantage PLUS kit and the Human Neurology 4-Plex E (N4PE+) Advantage Plus assay kit on the HD-X platform[19].

Statistical Analyses

Fisher's exact test was used for binary features and the Mann-Whitney U test was used for quantitative features to analyze the clinical features, grouped by cognitive stages. Continuous variables were summarized using means and standard deviations (SDs) and categorical data were expressed as counts and percentages. Univariate and multivariable logistic regression analyses were performed to identify factors associated with the occurrence of AEs. Variables with a P value < 0.1 in univariate logistic analysis, as determined by the likelihood ratio test, were entered into the final multivariable model. Sensitivity analysis was performed after excluding those with superficial siderosis and medium-sized infarcts at baseline. To estimate changes in amyloid centiloid, tau SUVR, and plasma biomarkers, linear mixed-effects models with random slopes and intercepts were adjusted for sex and age at first infusion. For cognitive assessments, the models were additionally adjusted for baseline severity of cognitive impairment, education, study center, and concomitant use of acetylcholinesterase inhibitors and memantine. False discovery rate (FDR) correction was applied for multiple testing. Statistical analyses were performed using R, version 4.4.1, with all tests two-sided at a 5% level of significance.

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Results

Demographics and Clinical Characteristics of Patients

A total of 407 AD patients initiating lecanemab infusion were included in the overall study sample. The mean follow-up time was 5.6±3.39 months (excluding withdrawals and dropouts). The number of patients who completed 3-month, 6-month, and 9-month follow-up visits, as well as dropouts, is shown in Fig. 1B. The number of patients diagnosed with mild cognitive impairment, mild dementia, and moderate dementia was 239, 136, and 32, respectively (Table 1). The mean age was 68.08 years, with 67.57% of patients being female. There were no significant differences in age and gender among the three groups. The mean education was 9.57 years (SD, 4.71), with the moderate dementia group having the lowest years of education (7.00 years [SD, 4.56]). The average course from symptom onset was 2.65 years (SD, 1.54), with the moderate dementia group sharing the longest disease course (4.12 years [SD, 1.88]). 56.51% of the patients were APOE ε4 carriers (43.98% heterozygous, 12.53% homozygous). Comorbidities of hypertension, diabetes, and dyslipidemia were observed in 32.68%, 13.51%, and 28.26% of the patients. Most patients were using symptomatic medications in combination with lecanemab (70.52% acetylcholinesterase inhibitor, 23.83% N-methyl-D-aspartic acid receptor antagonist). Three patients (0.74%) were receiving anticoagulant medications, and 35 patients (8.60%) were taking antiplatelet agents. Characteristics of each center are demonstrated in Supplemental Table 1.

Most patients underwent amyloid PET (378 [92.87%]) for confirmation of amyloid pathology, while twenty-three patients (5.65%) underwent CSF AD biomarker testing. Four patients (0.98%) conducted both PET and CSF examinations. Two patients (0.49%) only underwent a high-accuracy blood AD biomarker test for amyloid pathology. Notably, among 382 patients who underwent PET imaging, 243 patients also performed simultaneous tau PET for pathological staging of AD. Most patients in MCI (85.1%) and mild dementia (84.8%) group were at stage C, while 45.5% of the moderate dementia patients receiving lecanemab treatment were at stage C (Fig. 1C).

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Treatment-related Symptoms

Of the 407 patients, 2 were lost to follow-up regarding their most recent infusion. In the remaining 405 patients with available infusion data, 90 (22.22%) experienced treatment-related symptoms during the study period. Numbers and AEs in participants who completed 3-month, 6-month, and 9-month follow-up visits are shown in Supplemental Table 2. The actual number of patients finishing each infusion, with the percentage and severity of treatment-related symptoms, was demonstrated in Fig. 2A. Most treatment-related symptoms occurred after the first infusion, and the incidence rate sharply decreased after the second infusion. Symptoms were usually rated as mild or moderate, with only three severe cases reported. Except for general disorders normally caused by IRRs, the most affected system was nervous system disorders, followed by gastrointestinal disorders. Commonly reported symptoms included pyrexia, dizziness, chills, vomiting, and headache (Fig. 2B). Rarely reported symptoms of abnormal liver function indicators and sudden deafness were observed. Details of the rare symptoms were shown in the supplement. The three serious adverse reactions were sudden deafness, lower limb thrombosis, and severe symptomatic ARIA.

IRRs was reported in 41 patients (10.12%), with the most common symptoms being pyrexia and chills, and approximately half of these reactions (21/41) were mild. Most patients experienced IRRs in the first infusion, except for two patients whose infusion-related blood pressure changes occurred in subsequent infusions. Drug intervention was carried out in 23 patients with IRRs. 27 (65.85%) patients recovered within 24 hours (Fig. 2C). Of 37 patients who conducted at least one infusion after first IRR, nine patients (24.32%) experienced more than one instance of IRR, and eight of them experienced moderate IRRs in first infusion. Among the 14 patients whose first IRRs lasted for more than 24 hours, 50% experienced similar IRRs during subsequent treatment. In contrast, the recurrence rate was 7.4% among patients whose first IRR resolved within 24 hours (Fig. 2D).

At the end of the study period, seven patients had arranged prophylactic treatment after experienced IRRs. The characteristics, symptoms, and prophylactic treatment are listed in Supplemental Table 3. In general, oral antihistamine drug or glucocorticoid therapy was administered by the neurologists based on the symptoms and severity of IRRs. Of note, five

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patients with more complex and persistent symptoms still presented milder but similar IRR symptoms on the subsequent infusion.

Baseline MRI and ARIA Findings

Before starting lecanemab infusion, patients underwent MRI scans to evaluate their cerebrovascular risk. On baseline brain MRI, 81 (19.90%) patients had lacunar infarctions, including 6 (1.47%) with medium-sized infarcts. The majority of patients (85.02%) had evidence of white matter disease, with a Fazekas score of either 1 (238 [58.48%]), 2 (89 [21.87%]), or 3 (19 [4.67%]). A total of 130 patients (31.94%) reported baseline microhemorrhage and 3 (0.74%) patients had superficial siderosis (Table 1).

A total of 321 patients who reached the point of monitoring scans with baseline MRI and monitoring MRI available were included in the study assessing ARIA risk. 307, 250, and 146 patients completed monitoring MRI before the 5th, 7th, and 14th infusion, respectively. Among them, 39 patients (12.15%) developed at least one ARIA finding. 16 had ARIA-E with or without ARIA-H and 23 had isolated ARIA-H (Table 2 and Fig. 3A). The incidence of ARIA was 7.17%, 10.99%, and 10.13% in patients who completed 3-month, 6-month, and 9-month follow-up visits (Supplemental Table 2). Isolated ARIA-H increased in 6 months (3.6% in 2 months and 4.8% in 6 months) (Fig. 3B). Most ARIA were asymptomatic (35/39 [89.74%]) and were radiographically mild (23/39 [58.97%]). Concurrent ARIA-E and ARIA-H were present in 12 patients, with seven moderate and severe cases demonstrating concurrent ARIA-E and ARIA-H in the same brain region (Fig. 3C). For four symptomatic ARIA cases, three patients detected ARIA after urgent MRIs, and one patient was detected on routine monitoring MRI and reported aggravated symptoms after being contacted about the MRI results. Two mild symptomatic cases were dizziness with mild ARIA-H. A total of 11 urgent MRIs were ordered for patients with symptoms that may be associated with ARIA and three discovered ARIA.

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Factors Associated with Adverse Events

We further explored factors that may associated with the risk of AEs. By grouping patients by different APOE ε4 status and the severity of cognitive impairment, we observed that there was no higher risk of all AEs in moderate AD dementia patients, while the incidence of IRRs was higher (Moderate AD Dementia, 21.88%; Mild AD Dementia, 5.93%; MCI due to AD, 10.92%,) (Table 3). After univariable and multivariable linear regression models, we demonstrated that over one microhemorrhage at baseline was associated with higher rates of AEs (OR[95%CI]: 2.55[1.31-4.95], *P*=0.006) and IRRs (OR[95%CI]: 2.23[1.03-4.68], *P*=0.036). Usage of antiplatelet (OR[95%CI]: 2.82[1.02-7.55], *P*=0.039) was independently associated with risk of AEs. (Supplemental Table 4&5). Sensitivity analysis demonstrated similar results.

After grouping by different APOE ε4 status and the severity of cognitive impairment, there were no significant differences in the risk of ARIA among these groups (Table 2 and Supplemental Table 6). In univariable and multivariable linear regression analysis, compared with those without microhemorrhage, occurrence of microhemorrhages at baseline was associated with the risk of ARIA (Baseline 1 microhemorrhage: OR[95%CI]: 4.96[2.06-11.97], *P*<0.001; Baseline over 1 microhemorrhage: OR[95%CI]: 4.98[2.10-11.83], *P*<0.001). For isolated ARIA-H, baseline superficial siderosis is detrimental (OR[95%CI]: 15.68[1.17-385.74], *P*=0.040), while the occurrence decreased with longer disease course (OR[95%CI]: 0.63[0.39-0.92], *P*=0.033). However, a higher number of baseline microhemorrhages was not found to correlate with an increased risk of isolated ARIA-H. (Supplemental Table 7-9).

Change in Lecanemab Usage

During this study period, 38 patients (9.38%) withdrew from lecanemab treatment, 29 patients (7.16%) experienced short-term discontinuation, and two patients were lost to follow-up. These patients withdrew from lecanemab treatment for the following reasons: 10 for asymptomatic and two for symptomatic ARIA; nine for treatment-related symptoms; 15 for subjective reasons including concerns about side effects, dissatisfaction with the therapeutic effect (worsen cognition), economic reasons, and inconvenient transportation; and two for unrelated health

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issues. Of all withdrawn patients, 71.05% were female. Patients who experienced ARIA had a higher withdrawal rate (Fig. 4A). The moderate AD dementia group had a higher withdrawal rate (Fig. 4B), but most of them withdrew for subjective reasons, including worsened cognition leading to poor medication adherence (Table 3). No statistical difference was found in the percentage of withdrawal and treatment interruption due to AEs in different cognitive groups. Characteristics and symptoms of patients who withdrew due to AEs are listed in Supplemental Table 10. Four patients withdrew after the first infusion due to intolerance of IRRs and two patients withdrew for aggravation of AD-related symptoms. Of 29 patients with short-term discontinuation, most of them (14/29) involved temporary discontinuation of medication due to mild or moderate ARIA, and nine patients experienced irregular administrations for subjective reasons or unrelated health issues.

Among 39 patients experienced ARIA, after the first findings of ARIA, 16 patients continued treatment, 14 patients suspended administration after findings of ARIA, six patients withdrew for moderate or severe ARIA, and three patients withdrew for concerns about the progression of ARIA. In patients who continued treatment, 14 patients observed stable or mild progressive ARIA in the following monitoring MRI, and the other two patients experienced newly discovered ARIA-E or severe ARIA-H. Of patients withdrew or suspended treatment, monthly monitoring MRI demonstrated stable ARIA-H or relieved ARIA-E after interruption of infusion. In 10 patients already restarted treatment, only one patient was found to have experienced a worsening ARIA and withdrew from the treatment. Representative MRI images of the patients who experienced severe ARIA-E and restarted treatment after ARIA relief are shown in Supplemental Fig. 1.

Biomarker and Cognition Monitoring

We further explored the short-term treatment outcomes of lecanemab by evaluating changes in amyloid biomarkers and cognitive function in patients who received lecanemab infusion for over 6 months. Given the limited number of patients completing these tests, we first compared baseline characteristics and AEs between patients with and without tau PET and plasma biomarker assessments to evaluate generalizability (Supplemental Tables 11 and 12).

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In 39 patients who had completed baseline and 6 months amyloid PET CT, we observed a declining trend in the amyloid centiloid from baseline to 6 months (β[95% CI]: -26.60[-32.19, -21.02], *P*<0.001) by a linear mixed-effects model, adjusted for sex and age at the first infusion (Fig. 5A). Since some patients' amyloid PET were performed by PET MR, to carry out a thorough assessment of amyloid changes, the rate of amyloid SUVR changes were evaluated. Linear regression model demonstrated a 12.47% decrease in amyloid PET SUVR after 6 months of treatment (95%CI=[-14.85, -10.08], *P*<0.001, Supplemental Fig. 2). Supplemental Fig. 3 demonstrated a case whose amyloid deposit detected by PET was close to negative after six months of treatment. Among 26 patients with contemporaneous follow-up tau PET, an increased tau SUVR in the cerebral cortex including inferior temporal, entorhinal, and parahippocampal regions was demonstrated, and SUVR in the anterior cingulate regions decreased (Fig. 5B and Supplemental Table 13).

The decrease of plasma P-Tau217 was observed after 3 months of lecanemab treatment (β[95% CI]: -0.16[-0.27, -0.054], *P*=0.006), and the P-Tau217 level met a more significant decline after 6 months (β[95% CI]: -0.26[-0.36, -0.17], *P*<0.001). The P-Tau217/ Aβ42 levels also decreased after 3 months (β[95% CI]: -0.23[-0.36, -0.10], *P*=0.001) and 6 months (β[95% CI]: -0.29[-0.40, -0.17], *P*<0.001). Moreover, plasma GFAP levels demonstrated a significant decrease after 3 months (β[95% CI]: -24.55[-37.87, -11.17], *P*=0.001) and remained stable in 6 months (β[95% CI]: -22.31[-33.82, -10.84], *P*<0.001). There was no significant difference in plasma NfL and Aβ42/40 levels both after 3 months or 6 months (Fig. 5C and Supplemental Table 14).

In 182 patients who completed 6 months of treatment, about half of the patients had follow-up cognitive assessments (MMSE: 61.53% [112/182], CDR: 39.56% [72/182], ADL: 25.27% [46/182]). Changes in MMSE, CDR, and ADL was shown in Fig.5D-5F. To initially determine the changes in cognitive performance after 6 months of lecanemab treatment, we compared the baseline and follow-up scores of MMSE, CDR, and ADL by linear mixed-effects models. After adjusting, a stable cognitive function was demonstrated with no significant changes in MMSE, CDR, and ADL scale after FDR correction (Supplemental Table 15).

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Discussion

In this study, we established a real-world setting to evaluate the safety and short-term changes of biomarkers and cognition with lecanemab in the Chinese population. Chinese AD patients adhered to lecanemab treatment with a withdrawal rate of 9.38%. Patients selected for lecanemab administration based on rigorous criteria achieved favorable benefits, with a relatively low rate of serious AEs. Throughout the study period, 22.22% of the patients experienced treatment-related symptoms, and 12.15% developed at least one ARIA finding. APOE ε4 status was not related to AEs in the Chinese population, while a higher total number of microhemorrhages at baseline was associated with an increased risk of AEs, pointing to the need for closer monitoring. Except for higher rates of IRRs in the moderate dementia group compared with the MCI group, no significant differences in AEs were observed. However, these patients may discontinue treatment due to poor medication adherence resulting from worsen cognition. After 6 months of treatment, the patients showed favorable amyloid pathology changes with stable cognitive function, while tau pathology continued to progressed.

Compared with treated participants from the whole group and the Asian region of Clarity AD trial[4,9], our patients receiving lecanemab were slightly younger and more cognitively impaired, with lower MMSE scores. Age is a major determinant of AD prevalence in China, with a peak incidence in individuals aged 75 years and above[20]. A major reason for the younger group of patients involved was the payment method in China. Unlike the payment methods in Western countries, where most payments may be covered by medical insurance[21], Chinese patients have to pay for lecanemab entirely out of their own pockets. Young patients have a more urgent need to maintain good cognitive function for a longer period, while older patients tend to allocate their funds for life-improving methods such as nursing homes. The huge economic burden also restricts the use of lecanemab in economically disadvantaged areas. Moreover, adhering to a two-week hospital visit schedule is a certain challenge for patients with poor transportation. Advantages of feasibility in China is that patients can quickly go from diagnosis to administration. They are usually able to complete the diagnostic testing within two weeks of the initial visit, and start the treatment within one week after the diagnosis.

Although the study duration of 11 months was relatively short, the occurrence rates of treatment-related AEs (27.9% vs. 29.5%), ARIA-E (5.0% vs. 6.2%), and ARIA-H (10.9% vs. 14.4%) were

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consistent with Clarity AD Asian Region. In contrast to real-world studies in Western populations[22-24], the overall incidence of AEs in the Chinese cohort was lower, both for treatment-related symptoms and ARIA. However, a higher proportion of patients discontinued treatment for subjective reasons, leading to a withdrawal rate similar to that seen in Western populations. No death and only one macrohemorrhage were reported during the study period. Overall, lecanemab demonstrated good tolerability in Chinese AD patients.

In terms of symptoms and the percentage of treatment-related AEs, this study showed consistency with previous findings on anti-Aβ monoclonal antibodies[25]. Notably, rare symptoms, including abnormal liver function indicators and sudden deafness, were observed. However, since these symptoms were discovered between the two administration periods rather than during the administration process, we cannot determine the correlation between these symptoms and lecanemab, and the possibility of complications cannot be ruled out. IRR was the most frequently reported AEs during the initial infusion and were predominantly mild. Mild IRRs that resolved within 24 hours rarely recurred with subsequent infusions. In contrast, patients who experienced moderate IRRs lasting over 24 hours may require prophylactic treatment before lecanemab infusion. Choices of prophylactic treatment include antihistamine, acetaminophen, or glucocorticoid therapy[26]. Preventive measures were generally effective, though severe conditions requiring glucocorticoid therapy were more likely to recur despite prophylaxis.

The majority of ARIA findings were asymptomatic and usually present as isolated ARIA-E or ARIA-H. ARIA-H findings were also noted in some patients in the placebo arm in clinical trials[4,27]. This suggests that the neuroimaging manifestations observed may not be solely attributable to treatment-related effects, but could also reflect inherent disease characteristics or distinctive neurovascular responses. Findings on associations between baseline microhemorrhages and incidence of isolated ARIA-H support this idea. Interestingly, our study found that APOE ε4 carriers did not exhibit a higher risk of ARIA, despite the genotype being recognized as a major risk factor for ARIA[7,28]. This may indicate that the response to anti-amyloid treatment varies across ethnic groups and underlying gene mechanisms, with the Chinese population displaying a distinct susceptibility profile. As APOE ε4 carriers may have same or better response to amyloid-targeting therapies[29], with better tolerance, the Chinese population may better benefit from monoclonal treatment. In patients with a higher number of microhemorrhages at baseline, there was a strong association with the occurrence of AEs, and

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therefore vigilant MRI monitoring is encouraged for greater clinical benefit. Most patients who developed ARIA did not withdraw from treatment. Monthly monitoring MRI revealed that most ARIA-H were stable, and ARIA-E can be resolved. Concerned about the progression of ARIA, some patients temporarily suspended administration and resumed once ARIA stabilized or resolved. Despite this, these patients showed no further progression of ARIA on subsequent MRI. Whether the suspension may affect the efficacy needs to be further studied.

We aware that there are instances where lecanemab is initiated beyond the appropriate use recommendation, such as in patients with moderate dementia, anticoagulant use, and widespread cerebrovascular disease. These patients nonetheless initiated therapy, reflecting strong willingness to pursue treatment despite acknowledgment of potential AEs and uncertain efficacy. Notably, 45.5% of the moderate dementia patients were at Stage C of tau PET biological stage, indicating a mismatch between clinical and pathological staging. Possibly caused by less neurodegeneration[30], thus suggesting the possibility of lecanemab treatment. Although previous Aβ monoclonal antibody trials failed to show clinical benefit in mild to moderate AD[31], these drugs were terminated for failing to achieve the expected therapeutic effects. Lecanemab has not been studied for moderate dementia. Inclusion of patients outside the scope of the clinical trial do give strength to the dataset as it provides evidence of the safety and efficacy profile in a more lenient real-world sample rather than simply recapitulating the trial inclusion criteria. This is particularly important to clinicians counselling patients about treatment who may be on the cusp of moderate AD or who are taking anticoagulants. In our study, safety outcomes were comparable to those in MCI, apart from a higher rate of IRRs. While withdrawal rates due to AEs did not differ across different cognitive groups, the moderate AD dementia group was observed to have a higher withdrawal rate due to worsened cognition and intolerance of the regular administration. Given the small sample size and limited follow-up, our data do not provide sufficient evidence to evaluate efficacy in moderate AD dementia. It is still inappropriate to make a conclusion on whether these patients can be widely treated with anti-amyloid monoclonal antibodies and the external validity is relatively low.

In preliminary analyses on changes in biomarkers, we examined the changes in PET and plasma biomarkers after 6 months of treatment. As expected, a reduction in brain amyloid burden was observed at 6 months by PET imaging. Consistent with Clarity AD and TRAILBLAZER-ALZ[32], clinical trial of another marketed monoclonal antibody drug Donanemab, a decrease in plasma P-

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Tau217 and GFAP levels can be observed 3 months after lecanemab treatment and continued at 6 months. No significant change was observed for NfL levels. As clearance of NfL in humans is very slow, failure to observe changes in NfL was not a surprise. Changes in Aβ42/40 demonstrated a significant increase in the extended analysis of Clarity AD, but were not replicated in our study. A longer-term follow-up is needed to clearly establish the changes. Cortical tau SUVR, however, showed an increase, and current evidence remains limited regarding whether lecanemab slows tau accumulation compared to non-treated patients. No public data on the effect of lecanemab treatment on tau-PET outcomes is yet available. The tau results here suggested that lecanemab treatment may not reverse or halt tau progression, and the effect on P-Tau markers may not be extrapolated to be an effect on tau tangle pathology. With TRAILBLAZER-ALZ phase 2 results demonstrating a decreased tau load[33], the effect of amyloid plaque-targeting monoclonal antibodies on tau pathology is still pending.

Though over half of the patients had improved scores in MMSE after six months of treatment, there was no statistical difference in all cognitive assessments. The improving trend differed from the Clarity AD and real-world study results[22] which demonstrated a declining trend in cognitive assessment score. Practice effect on repeated assessments in this open-label setting may be one reason[34]. Moreover, some of the patients included were newly diagnosed with AD, and the initial use of symptomatic improvement drugs contributed to the improvement in cognitive function. Restricted by the limited sample size of follow-up biomarker testing and cognitive assessments, the correlation between clinical and biological improvement and factors associated with better biological improvement was not detected in this study. Further study on this population is expected to explore which subgroups will benefit the most from the application of lecanemab treatment.

This study represents the first real-world analysis of lecanemab usage across multiple regions of China, comprising a relatively large sample size and comprehensive documentation of complications. The data and results are expected to be used for policy-making, service-planning and meta-analyses in the future. However, several limitations must be acknowledged. First, the relatively short follow-up period and small sample size may lead to an underestimation of AEs and hinder the evaluation of changes in biomarker and cognition. These factors may also made it difficult to draw a solid conclusion, including the relation between APOE ε4 carriers and the risk of ARIA. Still, with ongoing follow-up, we expect updated data on a larger study group at the

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12-month and 18-month marks, which will be crucial for determining the long-term efficacy. Second, the lack of a non-treatment-controlled group limited our evaluation of the efficacy, analysis on biomarkers and cognition under an intention-to-treat framework are expected to better reveal the effect of lecanemab. Third, limited follow-up data on biomarkers and cognitive assessments have imposed restrictions on the generalizability of the findings. A comparison between the group with follow-up biomarkers and the group without follow-up biomarkers demonstrated that AEs, especially the incidence of ARIA, may affect their enthusiasm for biomarker monitoring. Finally, the western region of China was not included in this study, which limited the generalizability of the results.

In conclusion, our study provided a real-world analysis of lecanemab use in the Chinese population, offering valuable insights into its benefits. The rate of AEs was low and manageable, with favorable biomarker changes observed in the short term. Lecanemab administration was well-tolerated and safe in the moderate AD dementia population, providing a basis for future prospective studies. Further confirmation will require extended follow-up, and future large-scale studies are warranted.

Data availability

Anonymized participant data can be shared with qualified investigators on request to the corresponding author, from the date of publication onwards.

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Competing interests

The authors declare no conflicts of interest.

Supplementary material

Supplementary material is available at *Brain* online.

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Figure legends

Figure 1 Workflow of the study, number of follow-up patients, and biological stages of the included patients. (**A**) Workflow of patient inclusion and monitoring. (**B**) Flow of the number of patients who completed 3-month, 6-month, and 9-month follow-up visits. (**C**) The percentages of biological stages determined by tau PET in the MCI, mild AD dementia, and moderate AD dementia groups. Abbreviations: AD, Alzheimer's disease; ARIA, amyloid-related imaging abnormalities; CDR, Clinical Dementia Rating; CSF, cerebrospinal fluid; MCI, mild cognitive impairment; N, number of patients; PET, positron emission tomography.

Figure 2 Characteristics of treatment-related symptoms over the treatment period. (**A**) The percentage of patients with treatment-related symptoms in 405 patients who were not lost to follow-up. The upper number indicates the exact number of patients finishing each infusion. (**B**) The percentage of symptoms reported by patients with treatment-related symptoms is shown, with bar colors representing different organ systems categorized by systemic organ categories. (**C**) The Venn diagram shows the situations in which patients experienced their first IRR. The blue circle indicates patients who received medication treatment for IRRs. The pink circle represents patients who experienced IRR in the first infusion. The yellow circle indicates all patients who had experienced IRRs. The green circle represents IRRs that were alleviated within 24 hours. (**D**) Situations of IRRs in 37 patients who received at least one infusion after their first IRR. Abbreviations: IRR, infusion-related reaction; N, number of patients.

Figure 3 Characteristics of ARIA over the treatment period. The analyses of ARIA included 321 patients who received baseline MRI and at least one monitoring MRI scan. (**A**) The proportion of patients with any ARIA-E, isolated ARIA-H, or any ARIA is shown as a function of the time since first infusion. The numbers below indicate the population at risk at the current time point, and the numbers in parentheses represent the number of patients who had experienced the event. Vertical ticks on the curve represent patients censored at the longest follow-up time. (**B**) The percentage of ARIA events reported in the monitoring MRIs before the

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5th, 7th, and 14th infusion. (**C**) The number of cases is shown according to the maximum radiographic severity of ARIA-H and ARIA-E. The size of the circle corresponds to the number of patients. Abbreviations: ARIA, amyloid-related imaging abnormalities; ARIA-E, ARIA with edema/effusion; ARIA-H, ARIA with hemorrhage/hemosiderin deposition.

Figure 4 Time-to-withdrawal curves of the patients. (**A**) Time-to-withdrawal curves in all patients with follow-up (N=405), in patients who experienced treatment-related symptoms (N=90), and in patients who developed ARIA (N=39). (**B**) Time-to-withdrawal curves in MCI (N=238), mild AD dementia (N=135), and moderate AD dementia (N=32) groups. Abbreviations: AD, Alzheimer's disease; ARIA, amyloid-related imaging abnormalities; MCI, mild cognitive impairment.

Figure 5 Biomarkers and cognition changes between baseline and 6 months. (**A**) Amyloid centiloid value at baseline and 6 months after treatment. A linear mixed-effects model, adjusted for age at first infusion and sex, was applied to evaluate the differences. (**B**) The cortical and subcortical t-values of follow-up tau SUVR compared with baseline by a linear mixed-effects model, adjusted for age at first infusion and sex. (**C**) The results of plasma biomarker changes by a linear mixed-effects model, adjusted for age at first infusion and sex. (**D**) Paired box plots showing changes in MMSE between baseline and 6 months. (**E**) Paired box plots of ADL changes. (**F**) Alluvial plot illustrates categorical transitions in CDR-G between baseline and 6 months. ns, not significant; **, $P < 0.01$; ***, $P < 0.001$. Abbreviations: Aβ, β-amyloid; ADL, Activities of Daily Living; CDR-G, Clinical Dementia Rating Global Score; GFAP, glial fibrillary acidic protein; MMSE, Mini-Mental State Examination; N, number of patients; NfL, neurofilament light chain; SUVR, standardized uptake value ratio.

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1 **Table 1 Baseline characteristics of patients**

Characteristic	Total	MCI due to AD	Mild AD Dementia	Moderate AD Dementia	*P*
Number of patients	407	239	136	32	
Age at first infusion, y, mean (SD)	68.08 (8.71)	68.08 (8.65)	68.38 (8.39)	66.84 (10.57)	0.669
Female, *n* (%)	275 (67.57)	159 (66.53)	92 (67.65)	24 (75.00)	0.630
Education, y, mean (SD)	9.57 (4.71)	10.49 (4.46)	8.57 (4.79)	7.00 (4.56)	<0.001
Time since onset of symptoms, y, mean (SD)	2.65 (1.54)	2.38 (1.42)	2.76 (1.45)	4.12 (1.88)	<0.001
ApoE ε4 status, *n* (%)					0.024
ε4 Non-carrier	156 (38.33)	89 (37.24)	48 (35.29)	19 (59.38)	
ε4 Heterozygotes	179 (43.98)	110 (46.03)	62 (45.59)	7 (21.88)	
ε4 Homozygotes	51 (12.53)	32 (13.39)	17 (12.50)	2 (6.25)	
Unknown	21 (5.16)	8 (3.35)	9 (6.62)	4 (12.50)	
MMSE score, mean (SD)	20.20 (5.09)	22.73 (3.60)	17.52 (4.20)	12.17 (4.40)	<0.001
Concomitant medications, *n* (%)					
Acetylcholinesterase inhibitor	287 (70.52)	168 (70.29)	92 (67.65)	27 (84.38)	0.174
Memantine	97 (23.83)	33 (13.81)	45 (33.09)	19 (59.38)	<0.001
Antiplatelet	35 (8.60)	17 (7.11)	16 (11.76)	2 (6.25)	0.268
Anticoagulant	3 (0.74)	1 (0.42)	2 (1.47)	0 (0.00)	0.456
Baseline Fazekas score, *n* (%)					0.018
0	27 (6.63)	14 (5.86)	10 (7.35)	3 (9.38)	
1	238 (58.48)	150 (62.76)	72 (52.94)	16 (50.00)	
2	89 (21.87)	52 (21.76)	32 (23.53)	5 (15.62)	
3	19 (4.67)	6 (2.51)	12 (8.82)	1 (3.12)	
Unknown	34 (8.35)	17 (7.11)	10 (7.35)	7 (21.88)	
Baseline infarcts, *n* (%)					<0.001
≥1 Small infarcts (<10 mm)	75 (18.43)	49 (20.50)	23 (16.91)	3 (9.38)	
1 Medium-sized infarct (10–20mm)	6 (1.47)	2 (0.84)	1 (0.74)	3 (9.38)	
Baseline microhemorrhage number, *n* (%)					0.040
0	255 (62.65)	145 (60.67)	91 (66.91)	19 (59.38)	
1	66 (16.22)	45 (18.83)	19 (13.97)	2 (6.25)	
2–4	58 (14.25)	34 (14.23)	19 (13.97)	5 (15.62)	
5–9	6 (1.47)	4 (1.67)	2 (1.47)	0 (0.00)	
Unknown	22 (5.41)	11 (4.60)	5 (3.68)	6 (18.75)	
Baseline superfcial siderosis, *n* (%)	3 (0.74)	2 (0.84)	1 (0.74)	0 (0.00)	0.014

2 Fisher exact test was used for binary features and Mann-Whitney U test was used for quantitative features to analyze the clinical features.
3 Continuous variables were summarized using means and SDs and categorical data were expressed as counts and percentages. AD, Alzheimer's
4 disease; APOE, apolipoprotein E; MCI, mild cognitive impairment; MMSE, Mini-Mental State Examination; SD, standard deviation.
5

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Table 2 Occurrence of ARIA in different cognitive groups

	Total	MCI due to AD	Mild AD Dementia	Moderate AD Dementia	*P*
n	321	191	112	18	
Any ARIA, *n* (%)	39 (12.15)	23 (12.04)	14 (12.50)	2 (11.11)	0.984
Symptomatic ARIA, *n* (%)	4 (1.25)	1 (0.52)	3 (2.68)	0 (0.00)	0.234
ARIA-E, *n* (%)	16 (4.98)	10 (5.24)	5 (4.46)	1 (5.56)	0.950
Mild	10 (3.12)	7 (3.66)	3 (2.68)	0 (0.00)	
Moderate	4 (1.25)	2 (1.05)	1 (0.89)	1 (5.56)	
Severe	2 (0.62)	1 (0.52)	1 (0.89)	0 (0.00)	
Isolated ARIA-E, *n* (%)	4 (1.25)	4 (2.09)	0 (0.00)	0 (0.00)	0.252
Mild	2 (0.62)	2 (1.05)	0 (0.00)	0 (0.00)	
Moderate	1 (0.31)	1 (0.52)	0 (0.00)	0 (0.00)	
Severe	1 (0.31)	1 (0.52)	0 (0.00)	0 (0.00)	
ARIA-H, *n* (%)	35 (10.90)	19 (9.95)	14 (12.50)	2 (11.11)	0.789
Mild	23 (7.17)	15 (7.85)	6 (5.36)	2 (11.11)	
Moderate	7 (2.18)	2 (1.05)	5 (4.46)	0 (0.00)	
Severe	5 (1.56)	2 (1.05)	3 (2.68)	0 (0.00)	
Isolated ARIA-H, *n* (%)	23 (7.17)	13 (6.81)	9 (8.04)	1 (5.56)	0.889
Mild	16 (4.98)	10 (5.24)	5 (4.46)	1 (5.56)	
Moderate	4 (1.25)	2 (1.05)	2 (1.79)	0 (0.00)	
Severe	3 (0.93)	1 (0.52)	2 (1.79)	0 (0.00)	
Concurrent ARIA-E and ARIA-H, *n* (%)	12 (3.74)	6 (3.14)	5 (4.46)	1 (5.56)	0.772
Mild	5 (1.56)	5 (2.62)	0 (0.00)	0 (0.00)	
Moderate	4 (1.25)	0 (0.00)	3 (2.68)	1 (5.56)	
Severe	3 (0.93)	1 (0.52)	2 (1.79)	0 (0.00)	

Fisher exact test was used to analyze the clinical features. AD, Alzheimer's disease; ARIA, amyloid-related imaging abnormalities; ARIA-E, ARIA with edema/effusion; ARIA-H, ARIA with hemorrhage/hemosiderin deposition; MCI, mild cognitive impairment.

Table 3 Occurrence of adverse events and change in administration in different cognitive groups

	Total	MCI due to AD	Mild AD Dementia	Moderate AD Dementia	*P*
n	405	238	135	32	
Treatment-related AE, *n* (%)	113 (27.90)	72 (30.25)	30 (22.22)	11 (34.38)	0.175
Treatment-related AE (other than asymptomatic ARIA), *n* (%)	90 (22.22)	59 (24.79)	22 (16.30)	9 (28.12)	0.117
Infusion-related reactions, *n* (%)	41 (10.12)	26 (10.92)	8 (5.93)	7 (21.88)	0.022
Treatment adjustment, *n* (%)					
withdrawal from treatment	38 (9.38)	15 (6.30)	17 (12.59)	6 (18.75)	0.022
treatment interruption	29 (7.16)	20 (8.40)	7 (5.19)	2 (6.25)	0.500
AEs leading to treatment adjustment, *n* (%)					
leading to treatment withdrawal	21 (5.19)	10 (4.20)	9 (6.67)	2 (6.25)	0.564
leading to treatment interruption	20 (4.94)	13 (5.46)	5 (3.70)	2 (6.25)	0.707

Fisher exact test was used to analyze the clinical features.
Abbreviations: AD, Alzheimer's disease; AE, adverse events; ARIA, amyloid-related imaging abnormalities; MCI, mild cognitive impairment.

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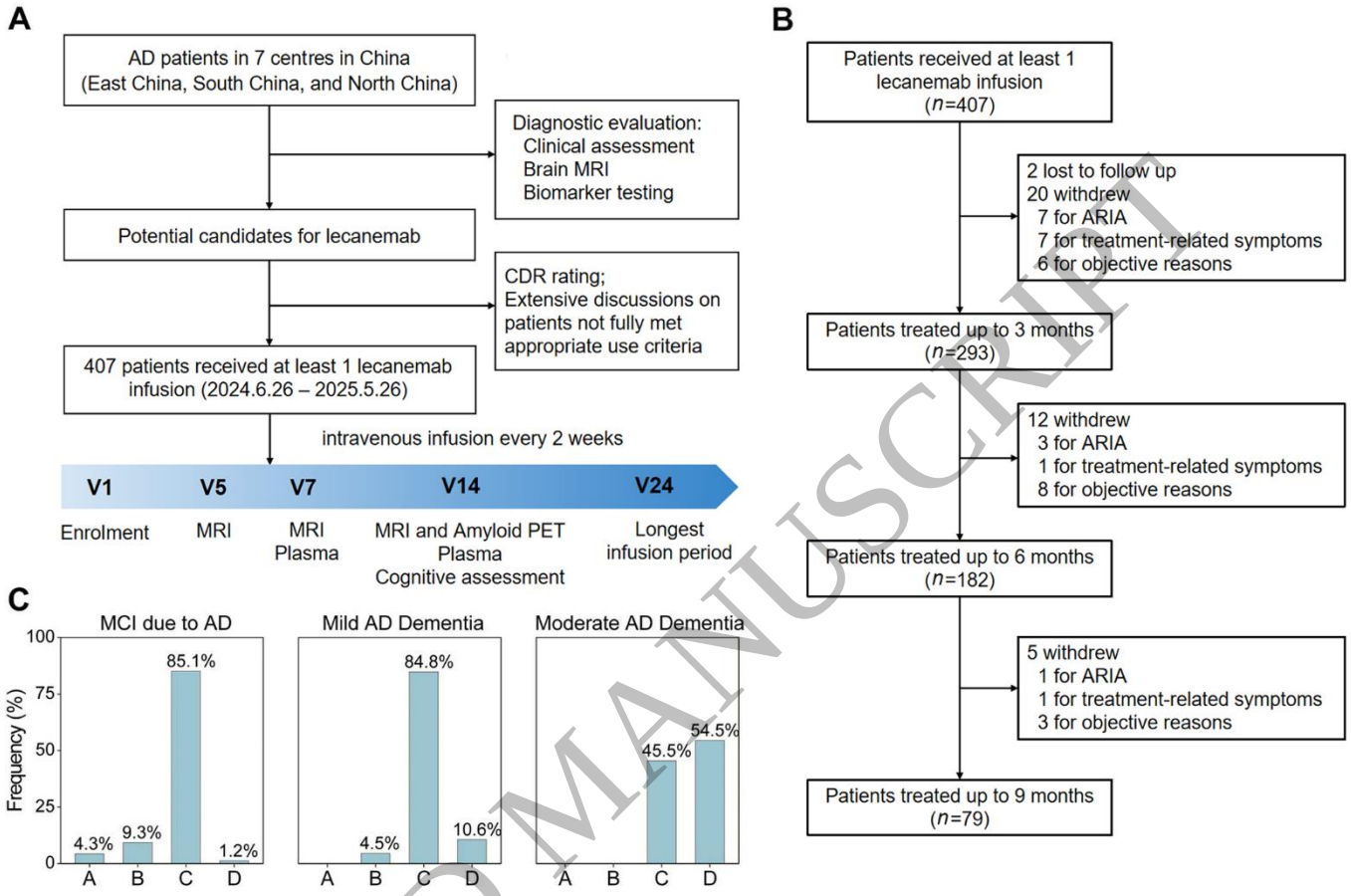

Figure 1
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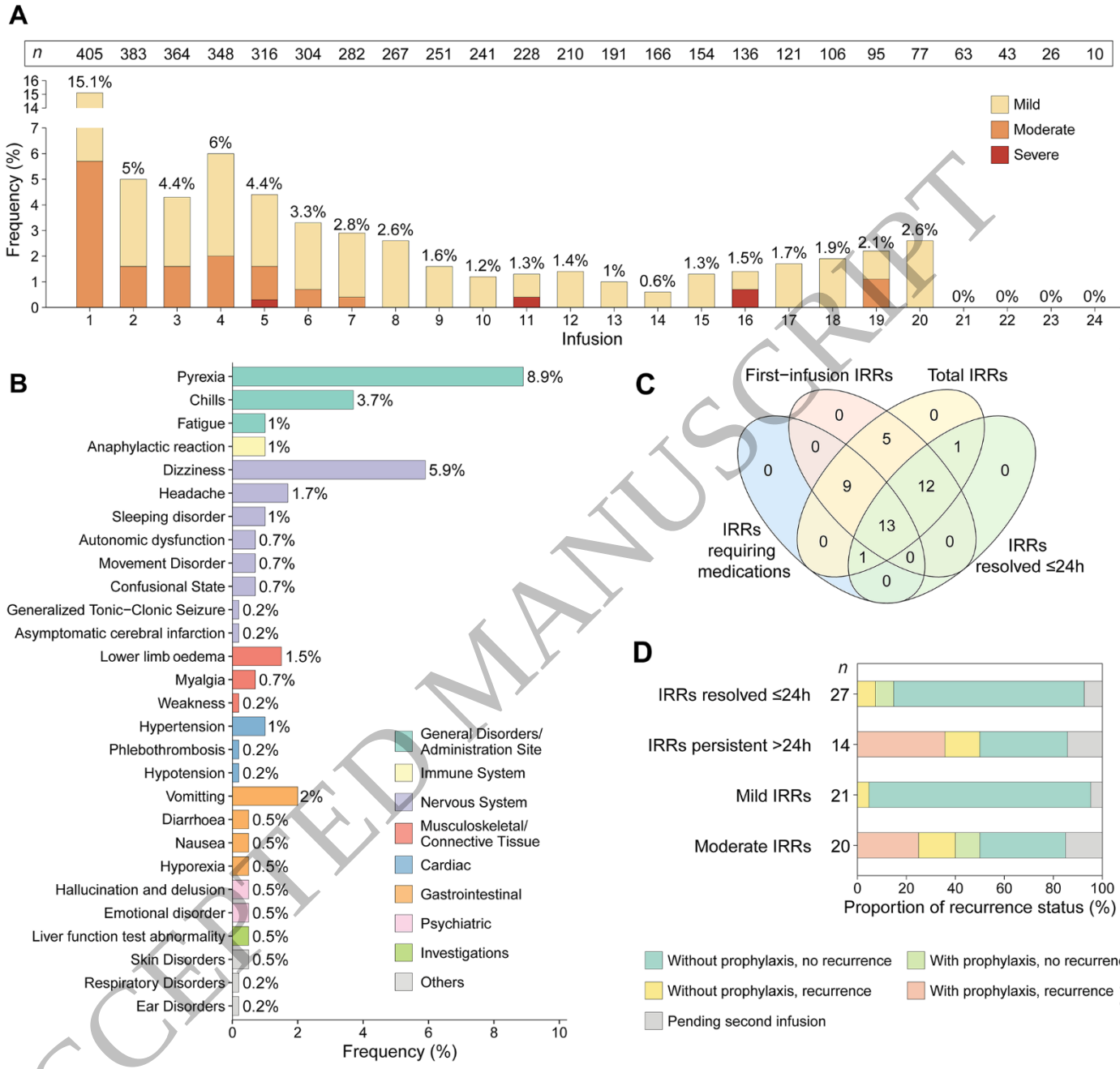

Figure 2
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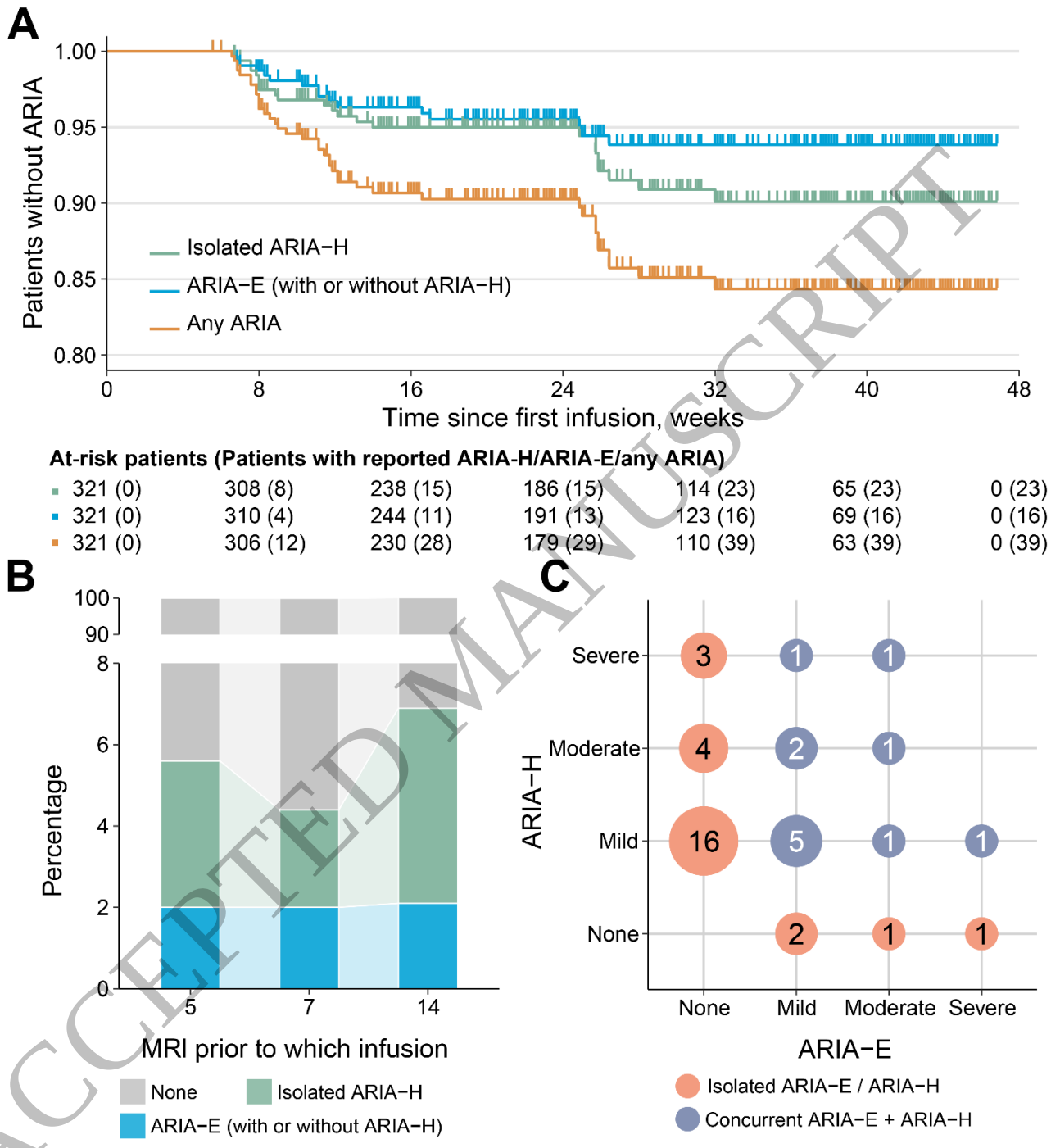

Figure 3
180x190 mm (x DPI)

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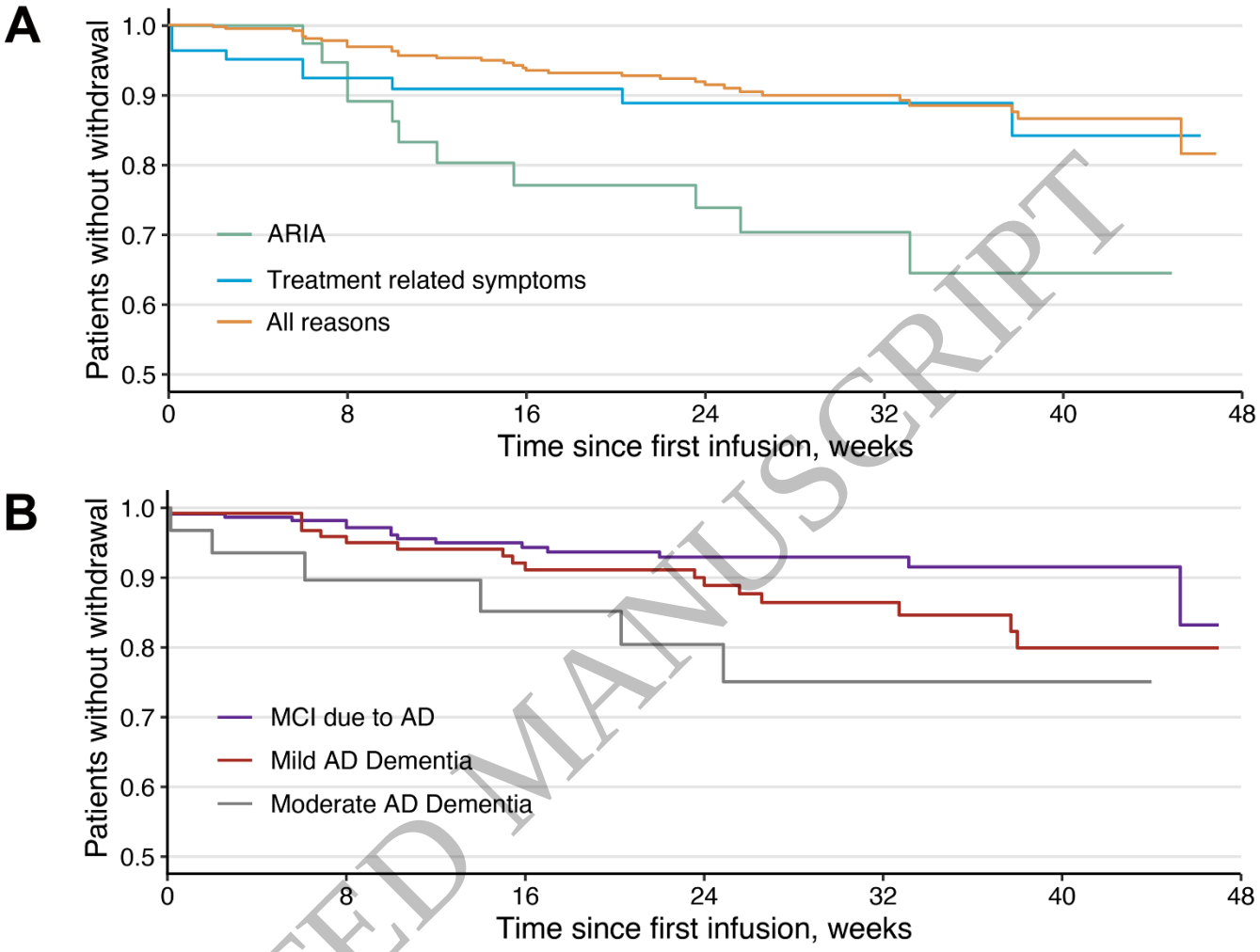

Figure 4
185x139 mm (x DPI)

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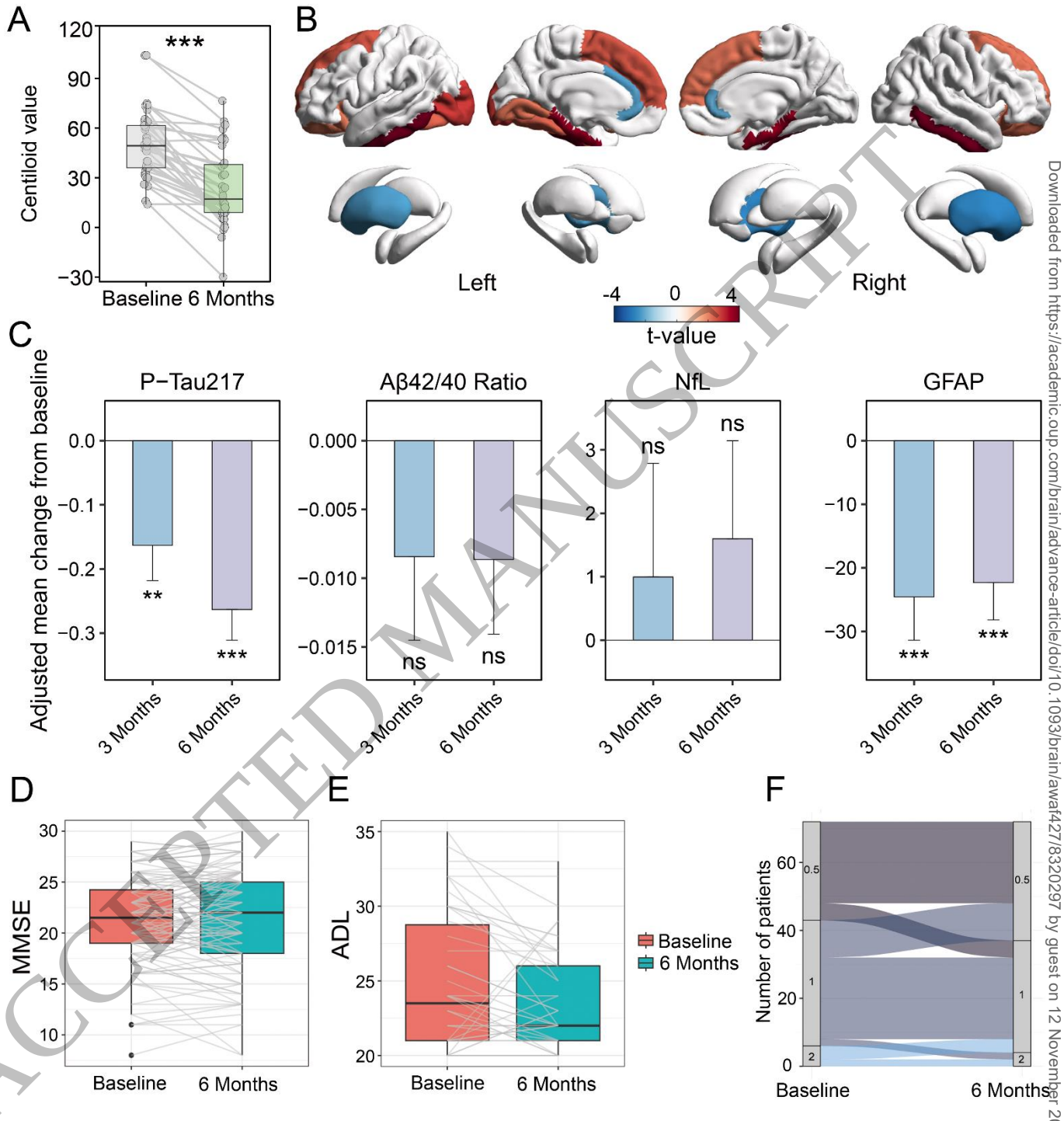

Figure 5
185x192 mm (x DPI)

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